

COURSE CODE: CSC214

COURSE TITLE: DATABASE MANAGEMENT SYSTEMS

SEMESTER: ALPHA

SESSION: 2021/2022

TEST 1: ATTEMPT ALL QUESTIONS

1a. Extensively differentiate between the following terms:

- i. Database Manager and Database Administrator
- ii. Data Manipulation Language and Data Definition Language
- iii. SQL and NOSQL Databases
- iv. Composite and Multi-valued Attributes

2. At a weekend retreat, the entity type PERSON has three subtypes: CAMPER, BIKER, and RUNNER. Draw a separate Extended Entity Relationship (EER) diagram segment for each of the following situations:

- a) At a given time, a person must be exactly one of these subtypes.
- b) A person may or may not be one of these subtypes. However, a person who is one of these subtypes cannot at the same time be one of the other subtypes.
- c) A person may or may not be one of these subtypes. On the other hand, a person may be any two (or even three) of these subtypes at the same time.
- d) At a given time, a person must be at least one of these subtype

3. Explain the following terms:

- a) Open Database Connectivity (ODBC)
- b) Data Mining
- c) Knowledge-Based Systems
- d) Locking Methods



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B.Sc DEGREE EXAMINATION

COLLEGE: SCEINCE AND TECHNOLOGY

DEPARTMENT: COMPUTER AND
INFORMATION SCIENCES

SESSION: 2021/2022

SEMESTER: ALPHA

COURSE CODE: CSC 214

CREDIT UNIT: 3

COURSE TITLE: HIGH PERFORMANCE AND DATABASE MANAGEMENT SYSTEM

INSTRUCTIONS: ANSWER ANY FOUR QUESTIONS

TIME: 3 HOURS

QUESTION ONE

Figure 1 represents the instance of the DreamHome rental database.

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyle St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manor Rd	Bristol	BS99 1NZ
B002	36 Clover Dr	London	NW10 6EU

Staff

staffNo	fname	lname	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Purd	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holthead	Aberdeen	AB2 3SU	House	6	650	CO46	SA9	B007
PL94	6 Argyle St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Client

clientNo	fname	lname	telNo	prefType	maxRent
CR76	John	Kay	0207-774-5632	Flat	425
CR56	Aline	Stewart	0141-848-1825	Flat	350
CR74	Mike	Ritchie	01475-392178	House	750
CR62	Mary	Tregear	01224-196720	Flat	600

PrivateOwner

ownerNo	fname	lname	address	telNo
CO46	Joe	Keogh	2 Pergus Dr, Aberdeen AB2 7SX	01224-361212
CO87	Carol	Ferrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

Viewing

clientNo	propertyNo	viewDate	comment
CR56	PA14	24-May-04	too small
CR76	PG4	20-Apr-04	too remote
CR56	PG4	26-May-04	
CR62	PA14	14-May-04	no dining room
CR56	PG36	28-Apr-04	

Registration

clientNo	branchNo	staffNo	dateJoined
CR76	B005	SL41	2-Jan-04
CR56	B003	SG37	11-Apr-03
CR74	B003	SG37	16-Nov-02
CR62	B007	SA9	7-Mar-03

- Write the DDL state to create PropertyForRent Table. (8.5 Marks)
- Write DDL statements to achieve the following; (4.5 Marks)
 - Delete property for rent table from the database
 - Change the Staff table by removing the default of 'Assistant' for the position column and setting the default for the sex column to female ('F').

- Spoke*
- iii. Change the PropertyForRent table by removing the constraint that staff are not allowed to handle more than 100 properties at a time. Change the Client table by adding a new column representing the preferred number of rooms
- (c) Write DML statements to achieve the following: **(4.5 Marks)**
- i. Find all staff whose salary is larger than the salary of every member of staff at branch B003
 - ii. List the properties that are handled by staff who work in the branch at '22 Deer Road'
 - iii. List full details of all branch

QUESTION TWO

(a) UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. create an Entity Relationship (E-R) diagram that captures this information about the UPS system. Be certain to indicate identifiers and cardinality constraints.

(9 marks)

(b) As the ability to handle large amounts of data improves, describe three business areas where these very large databases are being used efficiently. **(4.5 Marks)**

(c) Suppose you are given a relation R with four attributes ABCD. And functional dependency $C \rightarrow D$, $C \rightarrow A$, $B \rightarrow CF$, assuming those are the only dependencies that hold for R, do the following:

- i. Identify the candidate key(s) for R.
- ii. Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF).
- iii. If R is not in BCNF, decompose it into a set of BCNF relations that preserve the dependencies.

(4 Marks)

QUESTION THREE

Given the following problem scenario, presented as business rules;

Rehoboth housing has branch offices in various part of the country. The requirement specification for the company is stated below.

Each branch office is allocated members of staff and a manager to manage the operations of the office. The information to be held on the branch office includes a unique branch number, address (street, area ,city), telephone number and the name of the manager. A staff can only be allowed to manage a branch. There are special staff positions like manager, secretary and sales personnel. A staff can belong to more than one of the special positions. For each manager there is bonus, each secretary is characterized by typing speed and each sales rep. has sales area and car allowance A member of staff can either be on part-time (temporary staff) or full-time (permanent staff) depending on their appointment. Each full time staff is characterized by salary scale and hourly

2/2/22

allowance, while the part-time staff are rated per hour. The information stored on each member of staff includes staff number, name (first and last name), address, telephone number, sex, date of birth, position, salary and date of appointment. A staff number is unique across all branches of the company. When possible the details of the next-of-kin of staff members are stored. The information stored on each next-of-kin includes name, relationship to the member of staff, address and telephone number. Only the detail of one a single next-of-kin are held for a member of staff, and one next-of-kin cannot stand-in for more than two staff.

In addition, Rehoboth housing, has customers who attended to by one and only one member of staff, but a member of staff can attend to one or several customers. Figure 3a represents the customer rental form for the Rehoboth housing.

REHOBOTH

Page 1 **Customer Rental Details** Date 7 April 2009

Customer Name **Ibk Fat** Customer Number **RH 76**

Property Number	Property Address	Rent Start	Rent Finish	Rent	Owners Number	Owners Name
PG4	6, Lawrence St. Ikoyi	1-Jul-04	31-Aug-06	350	CO40	Tina Murphy
PG16	5, Novar St, Oshodi	1-Sep-06	1-Sep-08	450	CO93	Tony Shaw

Figure 2: Rehoboth Customer Detail form

- i. Develop the EER diagram for the problem scenario **(8Marks)**
- ii. Produce the logical design for the problem scenario **(7Marks)**
- iii. Where necessary, normalize the generated logical designs to 3NF **(2.5Marks)**

QUESTION FOUR

- (a) Model the following business rule based on supertype/subtype relationships. Show the constraints on the supertype/subtype relationships. List out clearly the attributes of each entity and model the supertype/subtype relationships.

Suppose you are asked to model the human resources in a university. Using specialization (a to-down approach) where the root general entity type is PERSON with attributes SSN (identifier), Name, Address, Gender and DOB. There are three types of persons that are of interest to the university which are: EMPLOYEE (persons who work for the university), STUDENT (persons who attend classes), and ALUMNUS (persons who have graduated). In this university a person might belong to more than one categories. Each employee has a value for Date hired and Salary and Student's Major department is recorded. For each ALUMNI the graduating degree is determined by year of graduation, designation and date of graduation. These two types of EMPLOYEE; FAULTY and STAFF have the following specific attributes respectively: Rank and Position. Also, a STUDENT could either be GRADUATE student with Test score or UNDERGRADUATE student with Class standing. (9.5 marks)

- (b) In the context of semantic object modelling, explain the following terms with diagrammatic illustrations
- i. Single/Multi-valued properties (2 marks)
 - ii. Object properties (2 marks)
 - iii. Compound objects (2 marks)
 - iv. Simple objects (2 marks)

QUESTION FIVE

- (a) Extensively differentiate between the following terms:
- i. Database Manager and Database Administrator (2 marks)
 - ii. SQL and NOSQL Databases (2 marks)
 - iii. Composite and Multi-valued Attributes (2 marks)
- (b) At a weekend retreat, the entity type PERSON has three subtypes: CAMPER, BIKER, and RUNNER. Draw a separate Extended Entity Relationship (EER) diagram segment for each of the following situations:
- i. At a given time, a person must be exactly one of these subtypes. (2 marks)
 - ii. A person may or may not be one of these subtypes. However, a person who is one of these subtypes cannot at the same time be one of the other subtypes. (2 marks)
 - iii. A person may or may not be one of these subtypes. On the other hand, a person may be any two (or even three) of these subtypes at the same time. (2 marks)
 - iv. At a given time, a person must be at least one of these subtype (2 marks)
- (c) What is an Enterprise-Level Database? Give a detailed list of its features and an accurate diagrammatic representation (3.5 marks)

QUESTION SIX

- (a) Draw a detailed Entity-Relationship diagram for the following business rule. Show the cardinality constraints for each relationship.

A Supplier may supply many Items. Each Item is supplied by any number of suppliers. Each Item must be used in the assembly of at least one Product and may be used in many products. A Supplier may send many Shipments. However, each Shipment must be sent by exactly one Supplier. A Shipment must include one or more Items. An item may be included on several

Shipments. A customer may submit any number of Orders. However, each Order must be submitted by exactly one Customer. An Order must request one or more products. A given Product may not be requested on any Order, or may be requested on one or more orders

(12 marks)

(2 marks)

(b) Mention four (4) characteristics of a good business rule

(c) An employee in an organization is represented by a unique ID, has a name which can be further broken down into first name and last name. The employee entity's date of employment is another valuable attribute, in addition to that, it has a payroll address which is comprised of the bank name, account number and sort code. The employee can have multiple skills and then the number of years employed can also be stored, although that can be gotten from the date employed. Represent this entity with an ER diagram showing the accurate representation of its attributes

(3.5 marks)